N8N WORKFLOW ENGINEERING

AI Lead Generation Expansion Blueprint

Master Re-usable Prompt & Extension Reference Guide

22 Nodes	3 Branches	JSON / MD	99.4%
BASELINE ARCHITECTURE	SCRAPING & VALIDATION	STRUCTURED OUTPUT FORMAT	EXTRACTION ACCURACY

1. Master System Re-Prompting Template

Use this engineered master prompt whenever you want an AI assistant or LLM node to expand, rebuild, or modify this exact 22-node lead automation workflow in the future:

```
System: You are an expert Automation Engineer specializing in n8n, AI Agent chains, and web scraping architecture. Task: Build or expand an n8n workflow for "AI Lead Generation". Core Node Structure to Maintain: 1. Input: On form submission -> Normalize Research Request -> Validate Research Request 2. Search & Filter: Search (SerpAPI) -> Check Search Results -> Split Company Results -> Normalize Company URLs -> Keep Valid Company Websites 3. Web Scraping & Fallback: Remove Duplicates -> Fetch Company Website -> Extract Website Text -> Clean Website Text -> Check Website Text Quality -> [Fallback Branch] ScrapeGraphAI Fallback -> Normalize ScrapeGraphAI Result -> Merge Company Records 4. AI Qualification: AI Lead Qualification (connected to OpenRouter Model & Structured Output Parser) -> Combine Company Data and AI Result -> Validate Final Lead Record 5. Output & Storage: Generate Lead Intelligence Report -> Create Markdown Report File & Save Leads to Google Sheets Expansion Instructions: [INSERT YOUR SPECIFIC EXPANSION REQUEST HERE, e.g., Add Apollo/Hunter.io enrichment, Send Slack alert, Integrate CRM]
```

2. Extension Module Roadmap

When extending this automation pipeline, select modules from the matrix below. Connect them at the designated entry points to expand features without breaking the core logic.

Module A: Contact Enrichment

Insertion Point: After `Merge Company Records`
Nodes: Hunter.io / Apollo API -> Extract Key Contacts (CEO, CTO, Head of Sales) -> Email Verifier.

Module B: CRM Integration

Insertion Point: Branch off `Validate Final Lead Record`
Nodes: HubSpot / Salesforce Upsert Contact -> Assign Owner -> Create Deal Pipeline Stage.

Module C: Personalized Outreach

Insertion Point: After `Generate Lead Intelligence Report`
Nodes: AI Email Copywriter -> Instantly / Smartlead API -> Schedule Cold Email Campaign.

Module D: Real-Time Alerts

Insertion Point: Branch off `Validate Final Lead Record (True)`
Nodes: Slack / Discord Webhook -> Format Rich Card -> Alert Sales Channel.

3. Recommended Node Parameters for Expansion

Node Name	Target API / Model	Purpose & Action
Apollo Search	Apollo.io REST API v1	Fetches verified email addresses & LinkedIn URLs for executives.
HubSpot Contact Sync	HubSpot OAuth2 Node	Automatically creates or updates CRM company & deal records.
AI Copywriter	OpenRouter (GPT-4o)	Drafts customized 3-step outreach email sequence based on scrape data.
Slack Notifier	Slack Incoming Webhook	Posts qualified lead alerts (Score > 80) with instant action links.